4.6 – Similarity and Transformations • Homework — Odd Answers

Name: _____

Find the image after a dilation centered at the origin with the given scale factor.

1. $(-2, 3)$, $k = 5$

Answer: $(-10, 15)$

2. $(12, -8)$, $k = 1/4$

Name the similarity transformation that maps the preimage onto the image.

3. $\triangle ABC$ with $A(1, 1)$, $B(2, 4)$, $C(4, 2) \rightarrow \triangle A'B'C'$ with $A'(3, 3)$, $B'(6, 12)$, $C'(12, 6)$

Answer: a dilation centered at the origin with $k = 3$

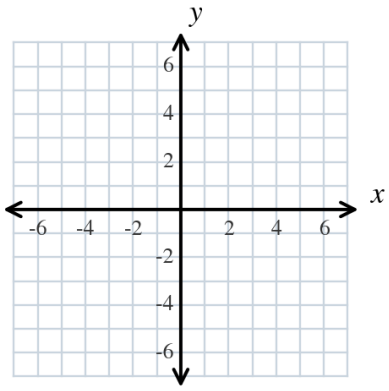
4. $\overline{DE}$ with $D(4, 6)$, $E(8, 2) \rightarrow \overline{D'E'}$ with $D'(2, 3)$, $E'(4, 1)$

Find the final image after each composition.

5. $(3, 1)$ after the translation $(x, y) \rightarrow (x - 2, y + 4)$, then a dilation about the origin with $k = 2$.

Answer: $(2, 10)$

6. $(8, -12)$ after a dilation about the origin with $k = 1/4$, then a reflection in the x-axis.



7. Graph $\triangle JKL$ with $J(1, 1)$, $K(2, 3)$, and $L(4, 1)$ and its image after a dilation about the origin with $k = 2$, then a reflection in the y-axis. List the final coordinates.

Answer: after the dilation: $J'(2, 2)$, $K'(4, 6)$, $L'(8, 2)$; after the reflection: $J''(-2, 2)$, $K''(-4, 6)$, $L''(-8, 2)$

Answer each question.

8. $\triangle MNP \sim \triangle XYZ$ with a scale factor of 4. If $MN = 2.5$ and $m\angle N = 65°$, find XY and $m\angle Y$.

9. Two figures are similar with $k = 1$. What else can you say about them?

Answer: they are congruent — $k = 1$ means no length changes, so the similarity transformation is a rigid motion

10. A student says every pair of similar figures must be the same size somewhere, because angles match. Explain the mistake.

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